# OM JEE

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## Projects

**Telehealth & Real-Time Consultation Engine** Project Link

*Node.js, Express.js, React.js, MongoDB, Socket.io, JWT, Redis, bcryptjs*

- Built a bidirectional messaging engine with Socket.io for real-time doctor–patient consultations.
- Designed a short-lived, stateful JWT guest-login flow enabling emergency consultation access without full registration.
- Built background worker scripts to periodically purge inactive guest profiles and stale chat history.
- Implemented real-time online/offline status and typing indicators by managing active socket connection state in memory.
- Used bcryptjs for password hashing and JWT for stateless role-based access control across Admin, Doctor, and Patient roles.

**Distributed URL Shortener & Analytics Engine** Project Link

*Node.js, Express.js, MySQL, Redis, Docker, Base62 Encoding*

- Built a horizontally scalable Node.js/Express.js backend supporting high-throughput concurrent traffic without local session dependency.
- Implemented short URLs using Redis atomic `INCR` for global IDs, encoded with a custom Base62 algorithm.
- Designed a cache-aside pattern with Redis for fast redirects, with fallback to MySQL during cache outages.
- Implemented hash-based sharding across MySQL instances, routing by short-code hash to avoid cross-shard joins.
- Deployed an IP-based sliding-window rate limiter backed by Redis to secure the write path against abuse.

## Technical Skills

**Languages:** C++, C, Python, JavaScript

**Web & Frameworks:** Node.js, Express.js, React.js, REST APIs, HTML5, CSS3

**Databases & Caching:** MySQL, MongoDB, Redis

**Real-Time & Authentication:** Socket.io (WebSockets), JWT, bcryptjs, Role-Based Access Control

**Tools & DevOps:** Docker, Git, GitHub

**Core Computer Science:** Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems

## Education

**The National Institute of Engineering (NIE), Mysuru** Sept 2023 – Jul 2027

B.E. in Information Science & Engineering **CGPA: 7.9**

## Achievements

- **Data Structures & Algorithms:** Solved problems on LeetCode and GeeksforGeeks, building consistent problem-solving proficiency.